

Tutorial–01: Fundamental Image Enhancement Techniques

Objective:

To implement and analyze basic image enhancement techniques in spatial and frequency domains using intensity transformations and histogram-based methods.

Experiment 1: Reading an Input Image

Theory: A digital image is represented as a two-dimensional matrix of pixel intensities.

Understanding size, data type, and histogram is essential.

Outcome: Ability to load, visualize, and interpret digital images.

Experiment 2: Image Negative

Theory: Image negation is a linear transformation that inverts pixel intensities using $s = 255 - r$.

Outcome: Understanding intensity transformations and contrast inversion.

Experiment 3: Log Transformation on Centered Fourier Spectrum

Theory: Fourier transform converts spatial data to frequency domain. Log scaling improves spectrum visibility.

Outcome: Understanding frequency-domain image representation.

Experiment 4: Gamma Correction on Bad Quality Image

Theory: Gamma correction adjusts brightness using $s = r^\gamma$. $\gamma < 1$ brightens, $\gamma > 1$ darkens images.

Outcome: Ability to enhance images using nonlinear transformations.

Experiment 5: Histogram Equalization

Theory: Histogram equalization redistributes intensity values to enhance contrast.

Outcome: Understanding histogram-based contrast enhancement.

Experiment 6: Histogram Specification

Theory: Histogram specification modifies an image to match a predefined histogram.

Outcome: Ability to perform controlled contrast enhancement

Experiment 7: CLAHE (Contrast Limited Adaptive Histogram Equalization)

Theory: CLAHE enhances image contrast by applying histogram equalization on small local regions while limiting contrast amplification to reduce noise.

Outcome: Understanding local contrast enhancement and noise control using adaptive histogram-based methods.

Final Outcome:

Students gain practical knowledge of fundamental image enhancement techniques and their applications.